

Saurabh Nichit

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Summary

Final-year Artificial Intelligence & Data Science undergraduate with strong foundations in Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, DBMS, and Software Development. Experienced in developing software applications using Python, C, C++, and Core Java, with hands-on experience in backend development, system-level programming, and AI-powered solutions. Passionate about solving real-world problems through scalable software engineering, continuous learning, and analytical thinking.

Technical Skills

Languages: C, C++ , Python, Core Java, SQL

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks, System Design Fundamentals.

Backend & APIs: FastAPI, Flask, RESTful APIs

Databases: MongoDB, MySQL, SQLite

AI & Machine Learning: Machine Learning, Deep Learning, TensorFlow, scikit-learn, OpenCV, Computer Vision, Natural Language Processing (NLP), LLMs, LangChain, RAG

Developer Tools:: Git, GitHub, Linux, Postman, Jupyter Notebook, VS Code.

Education

B.E. Artificial Intelligence & Data Science

2023-2027

PES Modern College of Engineering, Pune | **SPPU** | **CGPA: 9.14/10**

Higher Secondary Certificate (HSC)

2022

Swami Vivekanand Jr. College of Science, Commerce & Arts | **87%**

Secondary School Certificate (SSC)

2020

Sanskar Sarjan Education Society | **93.6%**

Experience

Software Developer Intern | Accurate Industries, Pune

Mar 2026–Jun 2026

[GitHub](#)

- Developed and maintained a full-stack industrial management platform by developing responsive web interfaces using HTML, CSS, and JavaScript while building scalable backend services with FastAPI and MySQL.
- Designed and implemented secure **RESTful APIs** for product management, customer inquiry processing, and administrative workflows, improving system functionality, application performance, and data accessibility.
- Built an interactive analytics dashboard to visualize customer inquiries, product performance, and operational insights, enabling data-driven decision-making through real-time reporting.
- Integrated an **LLM-powered chatbot** using **RAG** to provide contextual product assistance and automate customer support.
- Collaborated with stakeholders to gather business requirements, troubleshoot issues, optimize system performance, and delivered deployment-ready software following software engineering best practices

Projects

AI-Powered Dyslexia Screening & Handwriting Analysis Platform

[GitHub](#)

Python, Flask, TensorFlow, Scikit-learn, OpenCV, Tesseract OCR, SQLite, HTML, CSS, JavaScript

- Developed an AI-powered web platform for early dyslexia risk screening by combining cognitive assessment, handwriting analysis, and OCR-based text extraction into a unified application.
- Built a modular Flask backend with secure REST APIs, input validation, file upload handling, and AI inference pipelines for cognitive prediction and handwriting analysis.
- Integrated TensorFlow, Scikit-learn, OpenCV, and Tesseract OCR to implement handwriting analysis, cognitive risk prediction, and handwritten text extraction.
- Designed a responsive web interface with interactive assessment workflows, prediction dashboards, confidence analysis, and AI-generated recommendations.

Customized Virtual File System (CVFS)

C, Linux System Programming, Data Structures, File Handling

- Developed a Linux-inspired **Virtual File System (VFS)** in C with a custom command-line shell supporting file operations such as create, open, read, write, delete, and directory listing.
- Implemented core file system operations by simulating Linux system calls and designing internal data structures including the **Incore Inode Table**, **File Table**, and **User File Descriptor Table**.
- Designed a modular file management architecture with custom memory handling and file metadata management, providing practical exposure to operating system internals and low-level system programming.
- Built an interactive shell interface that emulates Linux file system commands, strengthening understanding of system calls, file handling, and operating system concepts.

Concurrent FTP Server

C, Linux, Socket Programming, TCP/IP, Process Management

- Developed a concurrent FTP server in C using **TCP/IP socket programming**, enabling multiple clients to connect simultaneously for reliable file transfer operations.
- Implemented core networking functionalities including client-server communication, file upload/download, request handling, and connection management using Linux socket APIs.
- Designed a multi-client architecture using **process-based concurrency (fork())**, ensuring independent client sessions and efficient resource utilization.
- Strengthened understanding of **Computer Networks**, **Inter-Process Communication (IPC)**, Linux system programming, and concurrent server design through practical implementation.

Generalised Data Structures Library

C++, Object-Oriented Programming (OOP), Templates, STL Concepts

- Developed a reusable **Data Structures Library** in C++ using templates to support multiple data types through generic programming, improving code reusability and maintainability.
- Implemented linear and non-linear data structures including **Linked Lists**, **Stacks**, **Queues**, **Binary Search Trees**, along with searching and sorting algorithms.
- Applied object-oriented programming principles such as **abstraction**, **encapsulation**, **inheritance**, and **polymorphism** to design a modular and extensible software library.
- Strengthened problem-solving skills by analyzing algorithmic complexity and implementing optimized data structure operations following software engineering best practices.

Leadership & Certifications

- **Department Head, NSS**: Leading AI & DS departmental activities; previously served as **Department Co-Head**, organizing volunteer initiatives and social outreach programs.
- **Class Representative (CR)**: Represented the AI & DS Department for **three consecutive years**, coordinating academic and departmental activities.
- **Code Club Member**: Active member of the AI & DS Department Code Club.
- **Smart India Hackathon (SIH)**: College-nominated team for an **AI-powered Integrated Waste Monitoring & Marketplace Platform**.
- **Advanced AI & Green Skills Certification**: Edunet Foundation, AICTE & Shell India.
- **Professional Certificate in Full Stack**: udemy